Enhancing Angular Sensitivity of Segmented Antineutrino Detectors

for Reactor Monitoring

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Localization of IBD Source

- Directional method for segmented Inverse Beta Decay (IBD) detectors
- Positron (e^+) gets most of the energy, and annihilation marks prompt vertex
- Neutron (^1n) gets momentum direction, and capture marks delayed vertex
- Lithium 6 dopant localizes ^1n capture with $^3_1$H and $^4_2$He depositing energy within $\sim 10\ \mu$m of capture
- direction to source inferred from ^1n direction with enough events

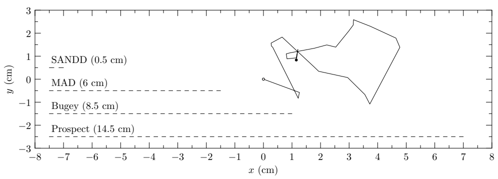


Figure 1: Diagram showcasing relative scale for characteristic neutron xy-trajectory projection from IBD vertex to capture location, as well as dimensions of segments in four 2D-segmented detectors: SANDD, MAD, Bugey, and PROSPECT.

Traditional Approach

- Conventional uncertainty calculation:

$$\Delta\varphi = \arctan\left(\frac{P/l}{\sqrt{N}}\right) \quad (1)$$

- If mean position resolution $P \to 0$, angular uncertainty $\Delta\varphi \to 0$ – unphysical!
- Vogel and Beacom model shows a large spread ($\sim 26°$) in intial neutron direction
- Neutrons scatter many times before capture

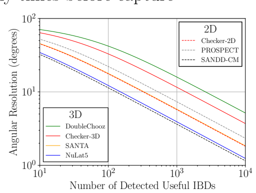


Figure 2: Angular uncertainty calculated using the conventional "Chooz" approach. A selected number of segmented detectors along with Chooz is highlighted. Useful IBD events are those that have prompt and delayed signals in separate segments. This result is taken from our previous study [?].

Our Approach to Simulation Design

- Used RATPAC-2 for all simulations
- Created simple monolithic geometry using 0.1%-wt ^{6}Li-doped plastic scintillator
- Created grid overlay to create binning matrices for 5 mm, 50 mm and 150 mm segment sizes in analysis work

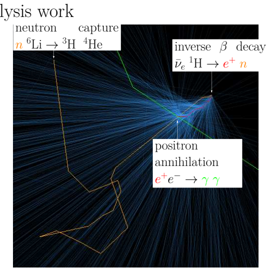


Figure 3: Rendering of RAT-PAC 2 IBD visualization. Positron track in red, neutron track — orange, gamma-rays — green, Cherenkov photons — cyan. Scintillation light turned off for clarity. Cherenkov photons indicate the direction of the positron, which traveled from the top right to the bottom left in this perspective (neutrino, not shown, is from top right to bottom left.

Novel Algorithm for Segmented Detectors

- Generate reference data for all incident angles
  This is done by rotating the grid overlay on simulation data and then performing the binning described below
- Bin prompt and delayed displacement vectors in matrix form by counting the number of voxels separating the IBD vertex and the capture vertex. Normalize all matrices for reference angles
- Create binning matrix for data set and normalize it
- Compute the L^2 norm for the matrix differences for all reference angles
- Fit $\alpha\left|\sin\left(\frac{\theta-\theta_0}{2}\right)\right|$ to L^2 norms as a function of angle. The minimum gives the θ_0 fit parameter for the direction of the neutrino source.

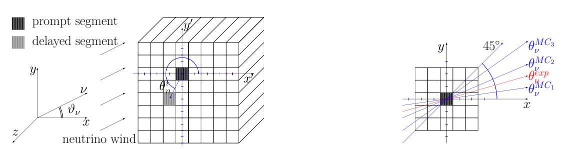


Figure 4: Diagram explaining the prompt-delayed segment geometry, as well as neutrino and axes orientation.

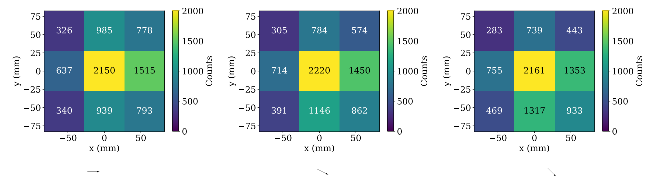
★ Conclusions, Findings, and Future Work

Key findings:

- At large n, the algorithm gives the direction of antineutrino sources
- At small n, the algorithm gives a reasonable uncertainty
- As a pattern matching method, the algorithm provides a comparison between distributions

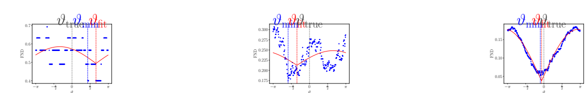
Future work:

- Developing methods to take advantage of machine learning
- Investigating how to apply to monolithic detectors
- Investigating a generalized approach to three dimensionally segmented detectors



$\vartheta_\nu = 0°$ $\vartheta_\nu = -30°$ $\vartheta_\nu = -45°$

Figure 5: Examples of binning distributions for incoming angles $\vartheta_\nu = 0°, -30°, -45°$, 0.1% ^{6}Li-doped. The direction of antineutrinos is indicated by the arrows below the plots.



(a) $n = 10$ (b) $n = 100$ (c) $n = 1000$

Figure 6: Explanation of angles and spread. The FND here is calculated from the RATPAC2 neutron capture data with a segment size of 50 mm.

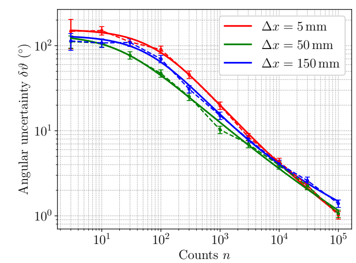


Figure 7: Angular uncertainty plot comparing the performance of three different square segment sizes: 5 mm, 50 mm and 150 mm.

References

[1] Crow *et al.*, in prep. (2026)

[2] Duvall *et al.*, Phys. Rev. Applied **22**, 054030 (2024)

Acknowledgments

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Ref. [1]: